Numerical Simulation of Quantized Vortices with Small Core Size via Vortex Tracking

Complex Quantum Systems and Vortex Dynamics
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1 Introduction

Ginzburg-Landau Hamiltonian with core size ε:

$$E_\varepsilon(u) = \int_\Omega \frac{1}{2}|\nabla u|^2 + \frac{1}{4\varepsilon^2}(1-|u|^2)^2 \tag{1}$$

- GP PDE: $i\partial_t \psi_\varepsilon = \Delta\psi_\varepsilon + \frac{1}{\varepsilon^2}(1-|\psi_\varepsilon|^2)\psi_\varepsilon$
- Ansatz for solution: $\phi_\varepsilon(x) = f_{\varepsilon,r_0}(|x|)\exp(i\theta(x))$
- $f_{\varepsilon,r_0}(0) = 0$, $f(\varepsilon, r_0)(r_0) = 1$
- draw graph

Renormalized energy

- Ginzburg-Landau energy diverges as $\varepsilon \to 0$.
- This divergence comes purely from the self-energy - no information about vortex interactions
- After subtracting the divergence we get the renormalized energy

$$W(a,d) = -\pi \sum_{1\le i\ne j\le N} d_i d_j \log|a_i - a_j| - \pi \sum_j d_j R(a_j; a, d) \tag{2}$$

- first term: vortex-vortex interaction
- second term: vortex-boundary interaction

- R is harmonic correction that enforces the Neumann boundary condition

Important Quantities:

- Supercurrents $j(\psi) = \frac{1}{2i}(\overline{\psi}\nabla - \psi\nabla\overline{\psi})$
- Vorticity: $J\psi = \frac{1}{2}\nabla \times j(\psi)$

For $\varepsilon \to 0$: $J\psi_\varepsilon(t) \to \pi \sum_{j=1}^{N} d_j \delta_{a_j(t)}$ The singular points solve:

$$\dot{a}_j(t) = -\frac{1}{\pi} d_j J \nabla_{a_j} W(a(t), d) \tag{3}$$

$$a_j(0) = a_j^0 \tag{4}$$

$$\nabla_{a_j} W(a, d) = -2\pi d_j \nabla_x (R(x; a, d) + \sum_{j \neq k}^{n} d_k \log|x - a_k|)|_{x=a_j} \tag{5}$$

Boundary terms in the renormalized energy $R(x; a, d)$ which solves

$$\Delta R = 0, \text{ in } \Omega \tag{6}$$

$$R = -\sum_{j=1}^{N} d_j \log|x - a_j|, \text{ on } \partial\Omega \tag{7}$$

- Following: condtions that determine the supercurrents
- They in turn determine harmonic map u^*

$$\nabla j(u^*) = 0 \tag{8}$$

$$\nabla \times j(u^*) = 2\pi \sum_{j=1}^{N} d_j \delta_{a_j} \tag{9}$$

$$\nu j(u^*) = 0 \text{ on } \partial\Omega \tag{10}$$

Where ν is the direction of the outward normal of $\partial\Omega$.
u^* is then given by

$$u^*(x; a, d) = \exp(iH(x)) \prod_{j=1}^{N} \frac{x - a_j}{|x - a_j|}^{d_j} \tag{11}$$

$H(x)$ is a harmonic function that is determined such that u^* satisfies the Neumann boundary condition.

Definition 1.1. A familiy of inital conditions $(\psi_\varepsilon^0)_{\varepsilon>0}$ is said to be well-prepared if it satisfies the following assumptions, for some constant $C >= 0$, $0 < \alpha < 1$ and ε small enough:

1. There exists N vortices with positions $a^0 = (a_j^0)_{j=1,\dots,N} \in \Omega^{*N}$ and degrees $d = (d_j)_{j=1,\dots,N} \in \{\pm 1\}^N$ such that

$$||J\psi_\varepsilon^0 - \pi \sum_{j=1}^N d_j \delta_{a_j^0}|| \le C\varepsilon^\alpha \tag{12}$$

   and the vortices are distant enough

2. the energy of ψ_ε^0 is close to be optimal:

$$E_\varepsilon(\psi_\varepsilon^0) \le W_\varepsilon(a_0, d) + C\varepsilon^{1/2} \tag{13}$$

$\dot{W}^{-1,1}$ is the dual space of $W^{1,\infty}$ The norm is given by:

$$||\mu||_{\dot{W}^{-1,1}} = \sup\{\int_\Omega \varphi \,\mathrm{d}\mu : ||\nabla\varphi||_{L^\infty} \le 1, \varphi \in W_0^{1,\infty}(\Omega)\} \tag{14}$$

Theorem 1.2. *Let ψ_ε solve the equation corresponding to the Schrödinger flow of the Ginzburg energy, with well-prepared initial conditions for some initial vortices with positions a^0 and degrees d_j. Then there exists $\gamma < 1$ and $C > 0$, such that for any $\varepsilon < \varepsilon_0$, well-preparedness is preserved along time, In particular,*

$$||J\psi_\varepsilon(t) - \pi \sum_{j=1}^N d_j \delta_{a_j(t)}||_{\dot{W}^{-1,1}} \le C\varepsilon^\gamma \tag{15}$$

$$||j(\psi_\varepsilon(t)) - j(u^*(a(t), d))_{L^{4/3}} \le C\varepsilon^\gamma \tag{16}$$

- Question: How do we construct a well-prepared family?

For ε, there is a unique minimizer $\phi_{\varepsilon,r_0}(x) = f_{\varepsilon,r_0}(|x|)\exp(i\theta(x))$ where f_{ε,r^0} satisfies an ODE, which we have seen before.
Starting from the localized approximation ϕ_ε of a single vortex one can the build the following inital condition:

$$\psi_\varepsilon^*(x; a^0, d) = u^*(x; a^0, d) \prod_{j=1}^N f_\varepsilon(|x - a_j^0|), \quad x \in \overline{\Omega} \tag{17}$$

which has the well preparedness properties.

2 Method

Idea

- FEM requires fine mesh for small ε
- here $\varepsilon isnotthelimitingfactor$

- we can use the point vortex system to track the vortex positions
- we can then build an approximation of the wave function from the vortex positions and use it as an initial condition for the PDE solver

R can be solved numerically like this:

1. Choose a maximum degree n and fix it
2. Compute the Fourier coefficients of the Dirichlet boundary condition up to order n for instance using a FFT
3. Compute the approximate solution R_n as the harmonic expansion of $\mathbb{P}_n g_a$

$$R_n(r\exp(i\theta)) = \sum_{k=-n}^{n} r^{|k|}\hat{g}_a(k)\exp(ik\theta) \tag{18}$$

2.1 Algorithm

2.1.1 Step 1: Constructing well-prepared initial conditions

- Define initial vortex positions and degrees. Set up the initial phase H such that the Neumann boundary condition is satisfied.
- Compute the radial function f_ε

2.1.2 Step 2: evolve vortex positions via Hamiltonian ODE

solve the ODE using RK4 and compute R as described before.

2.1.3 Computing R

- The solution R is harmonic and the boundary conditions are smooth as long as the vortices stay away from the boundary.
- We use harmonic polynomials as a basis to solve the PDE.
- For simplicity we choose as the unit disk.

harmonic polynomials:

$$\begin{cases} h_1 = 1 \\ h_{2m} = \Re(x+iy)^m \\ h_{2m+1} = \Im(x+iy)^m \end{cases} \tag{19}$$

for $x, y \in \mathbb{R}$.

- When Ω is the unit disk, the restriction of these polynomials to the boundary is nothing else than the Fourier basis.
- Denote by $\mathbb{P}_n$ the orthogonal projection from $L^2\partial(\Omega)$ to $\mathcal{X}_n := \text{span}\{h_k \mid_{\partial\Omega}, i \leq k \leq 2n+1\}$.
- In polar coordinates: $\Delta R = \frac{1}{r}\partial_r(r\partial_r R) + \frac{1}{r^2}\partial_\theta^2 R$
- Ansatz: $R(r,\theta) = f(r)\exp(ik\theta)$ for some $k \in \mathbb{Z}$.
- This gives us the ODE: $f''(r) + \frac{1}{r}f'(r) - \frac{k^2}{r^2}f(r) = 0$
- Using the Ansatz $f(r) = r^\alpha$ you get $\alpha = \pm|k|$
- one gets: $R_k(r,\theta) = r^{|k|}\exp(ik\theta)$
- PDE is linear therefore linear combinations are solutions
- on the unit circle for harmonic functions one gets: $R(r,\theta) = \sum_{-\infty}^{\infty} c_k r^{|k|}\exp(ik\theta)$
- the coefficients are determined by the boundary conditions
- At the boundary: $R(1,\theta) = \sum_{-\infty}^{\infty} c_k \exp(ik\theta) = g(\theta)$
- this is just the Fourier expansion of g
- therefore $c_k = \frac{1}{2\pi}\int_0^{2\pi} g(\theta)\exp(-ik\theta)\,\mathrm{d}\theta$

2.1.4 Step 3: Smoothing

At time $t > 0$ build back an approximation of the wave function from the vortex positions $a(t)$ as

$$u^*(x; a(t), d) = \exp(iH(x))\prod_{j=1}^{N}\left(\frac{x-a_j}{|x-a_j|}\right)^{d_j} \tag{20}$$

- u^* has singularities at the vortex locations a_j
- This is the reason we need to smooth it out before using it as an initial condition for the PDE
- Key insight: ϵ only enters through the 1D preprocessing - it is no longer a discretization bottleneck
- H solves: $\Delta H = 0$ in Ω and $H = -\sum_{j=1}^{N} d_j\partial_\tau \log|x-a_j|$ on $\partial\Omega$
- τ is the direction of the tangent to $\partial\Omega$
- H can also be solved using the same harmonic polynomial basis

3 Error Analysis

Lemma 3.1. *Let $\Omega \subset \mathbb{R}^2$ be the unit disk, let $\{a(t)\}_{t\geq 0}$ denote the exact solution to the Hamiltonian mechanics. Let $b(t)$ be the approximated ODE trajectory with the same degrees and initial positions b^0 with harmonic polynomials of degree up to n and time step δ_t. Suppose that for some $T > 0$,*

$$\rho_T = \min_{t\leq T}\min_{j\neq k}\{|b_j(t) - b_k(t)|, dist(b_k(t), \partial\Omega), |a_j(t) - a_k(t)|, dist(a_j(t), \partial\Omega)\} > 0 \tag{21}$$

is such that $0 < \rho_T < 1$. Then we have:

$$|a(t) - b(t)| \leq C(|a^0 - b^0| + (1-\rho_T)^n n^{1-l}) + C\delta t^4 \tag{22}$$

In order to prove this we need 2 other lemmas

Lemma 3.2. *Let $b^0, a^0 \in \Omega^{*N}$, for fixed $d \in \{\pm 1\}^N$ and let $\rho(a) := \min_{j\neq l}\{dist(a_j, \partial\Omega), |a_j - a_l|\}$. Suppose that $\rho_T := inf_{t\in[0,T]} \leq 0$*

$$|a(t) - b(t)| \leq C(|a^0 - b^0| + \frac{sup_{s\in[0,T]}||\mathbb{P}_n^{\perp} g_{a(s)}||_{L^2(\partial\Omega)}}{\rho_T^{3/2}}) \exp Ct/\rho_T^{5/2} \tag{23}$$

Where $g_a(x) = -\sum_j^N d_j \log|x - a_j|$

Proof.

- $|F^n(b) - F(a)| \leq E_1 + E_2 + E_3$
- $E_1 = \sum_j^N |\nabla_x R^n(b_j; b, d) - \nabla_x R^n(a_j; a, d)|$
- $E_2 = \sum_j^N |\nabla_x R^n(a_j; a, d) - \nabla R(a_j; a, d)|$
- $E_3 = \sum_{j\neq k} |\frac{b_j - b_k}{|b_j - b_k|^2} - \frac{a_j - a_k}{|a_j - a_k|^2}|$
- E_1 is the error made takinig different evaluation points on the boundary, E_2 discretizatio error, E_3 is the error made by the vortex-vortex interaction term

E_3 **error bound**

- E_3 is the error made by the vortex-vortex interaction term
- $|\nabla \log|x - a| - \nabla \log|x - b|| \leq C\frac{|a-b|}{\min\{|x-a|, |x-b|\}}^2$
- set $x = a_j$, $a = a_k$, $b = b_k$: $E_3 \leq C\frac{a-b||}{\min\{\rho(a), \rho(b)\}^2}$
- $\rho(a) = \min_{j\neq k}\{\text{dist}(a_j, \partial\Omega), |a_j - a_k|\}$

E_2 **error bound**

- discretization error
- the difference $R^n - R$ solve the Laplace equation with boundary data $P_n^{\perp} g_a$

- $\mathbb{P}_n^{\perp}$ is the orthogonal complement of $\mathbb{P}_n$
- this is everything that the fourier expansion of g_a up to order n does not capture $P_n^{\perp} g_a = g_a - P_n g_a = g_a - \sum_{k=-n}^{n} \hat{g}_a(k) \exp(ik\theta)$
- $\Delta(R^n - R) = 0$ in Ω, $R^n - R = P_n^{\perp} g_a$ on $\partial\Omega$
- One can show that $E_2(a) \leq \frac{||\mathbb{P}_n^{\perp} g_a||_{L^2(\partial\Omega)}}{\min_j \{\text{dist}(a_j, \partial\Omega)\}^{\frac{3}{2}}}$

E_1 **error bound**

- $\sum_j^N \nabla_x R_n(a_j; b, d)$ and using triangle inequality
- $E_1(a, b) \leq \frac{|a-b|}{\min_j \{\text{dist}(a_j, \partial\Omega), \text{dist}(b_j, \partial\Omega)\}}$

Putting everything together we get:

$$|\dot{a}(t) - \dot{b}(t)| \leq C(\frac{|a(t) - b(t)|}{\min\{\rho(a(t)), \rho(b(t))\}^2} + \frac{||\mathbb{P}_n^{\perp} g_{a(t)}||_{L^2(\partial\Omega)}}{\min_j \{\text{dist}(a_j, \partial\Omega)\}^{\frac{3}{2}}}) \tag{24}$$

- Using the assumption on ρ_T we get:

$$|\dot{a}(t) - \dot{b}(t)| \leq C(\frac{|a(t) - b(t)|}{\rho_T^2} + \frac{||\mathbb{P}_n^{\perp} g_{a(t)}||_{L^2(\partial\Omega)}}{\rho_T^{3/2}}) \tag{25}$$

- Inequality: $\frac{d}{dt}|a - b| \leq |\dot{a} - \dot{b}|$
- Proof of inequality: $\frac{d}{dt}|a - b| = \frac{d}{dt}\sqrt{(a - b) \cdot (a - b)} = \frac{(a-b)\cdot(\dot{a}-\dot{b})}{|a-b|}$
- use Cauchy Schwarz: $(a - b) \cdot (\dot{a} - \dot{b}) \leq |a - b| \cdot |\dot{a} - \dot{b}|$
- Gronwall-inequality: if

$$\frac{d}{dt} y(t) \leq f(t)y(t) + g(t), \tag{26}$$

  then $y(t) \leq y(0) \exp(\int_0^t f(s)ds) + \int_0^t g(s) \exp(\int_s^t f(\tau)d\tau)ds$

- Using Gronwall's inequality we get the desired bound

□

The time integration errors follows from the previous bounds and standard considerations on the numerical integration of ODEs

Lemma 3.3. *Let b be the exact solution to $\dot{b} = F^n(b)$ and let $(b_{\delta t}^k)_{k\delta t \leq T}$ be its numerical approximation with RK4 method and time step δt*

$$b_{\delta t}^0 = b^0 \tag{27}$$

$$b_{\delta t}^0 = b_{\delta t}^{k-1} + \delta t \Phi_{\delta t, n}(b_{\delta t}^{k-1}) \tag{28}$$

Where $\Phi_{\delta t,n}$ is the RK4 increment function associated with the approximate forcing term F^n. Assume that:

$$0 < \rho_T = \min(\inf_{0\leq t \ \ leqT} \rho(b(t)), \min_{k\delta t\leq T} \rho(b^k_{\delta t})) \tag{29}$$

Then there exists constants $C_{\rho_T} > 0$ and Λ_{ρ_T}, that depends on ρ_T but not on n, b or δt such that the following in time error holds:

$$\max_{k\delta t<T} |b(k\delta t) - b^k_{delta_t}| \leq \delta t^4 \frac{C_{\rho_T}}{\Lambda_{\rho_T}}(\exp(\Lambda_{\rho_T} T) - 1) \tag{30}$$

4 Numerical Results

- Show error increases with bigger ε
- Runtime: few seconds on laptop for $\varepsilon = 0.01$ and $N = 10$ vortices
- FEM solver took about a week to run (non-optimized code on a small cluster)
- Parameters of simulation: mesh size: $2 \cdot 10^{-3}$, time step: $\delta t = 10^{-5}$